



EXP2-1 Same or different Report P3

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Section 1: Identifying and correcting mistakes made during the experiment and data processing.

- **Type 1: Problematic recording of data;** Relevant group(s): P24; P40.

For P24, the recording of thickness is wrong (magnitude of stress reached 9.5MPa), the actual thickness of the sample should be 3 mm instead of 0.3 mm. For the second data point of P40, the length after elongation is 67mm while the initial length (no weight applied) is 67.2mm, strain is negative when stress is applied, which is impossible.

- **Type 2: No recording of initial distance of the specimen;** Relevant group(s): P38; P41

There is no recording of the initial distance between marks, which will not significantly affect the final modulus, but it is a basic experimental manipulation required in instruction. The new zero point was added, and new moduli were calculated.

- **Type 3: Wrong understanding of the experiment principle;** Relevant group(s): M10; MB.

The basic principle of this experiment is adding successive extra weights, recording successive weight and final distance. These two groups recorded the separate weight instead of the successive weight, so the stresses and calculated moduli are wrong. For correction, separate weight is accumulated, and a new stress-strain curve was constructed with new modulus.

- **Type 4: Exceeding elastic limit;** Relevant group(s): P33

When a large amount of stress is applied, the elastic limit of this specimen may be exceeded, with the presence of some dots that do not conform to the linear distribution. As the specimen is an elastomer whose stress-strain behavior should be first, elastic region where strain is proportional to the stress. Then strain-induced crystallization where the polymer chains are aligned, forming crystalline regions, leading to increased modulus. Once elastic limit is exceeded, the properties of the specimen have changed, and such data cannot be accounted as modulus. After deleting the last point, its modulus changed about 10% and the mean, SD of the sample also changed 0.2%, 0.4% (large for the sample) so it is decided to finally delete the last point.

Section 2: Data gotten are representative after correcting mistakes.

Parameter 1: R^2 In most groups, the R^2 values are close to 1, which shows the linear relationships between stress and strain measured in most groups are strong and can be fitted with a first-order function. It is further confirmed that the slopes can better represent the moduli of the specimens. **Parameter 2: Standard Error & Standard deviation;** In Unmodified P&M samples, standard errors are low (about 5% of the mean) which shows that sample means are closely distributed around the population mean and representative of the population. In modified P&M samples, the standard deviation was also small.

Section 3: Other relevant analyses.

1. Error analysis:

- Cause 1: Accuracy of measuring tools; Relevant groups: P2; P3; P7; P8; P15; P16; P20; P21; P25; P26; P28; P29; M2; M14; M16; M20; M30; M31; M36; M38; M39.

When a small amount of stress is applied, the strain remains the same. This may be because the strain is too small, and the measuring tool (meterstick) could not meet the measuring accuracy needed.

- Cause 2: Uneven internal mass distribution; Relevant group(s): Almost occur in every specimen.

The uneven internal mass distribution of the specimen (e.g. bubbles) leads to different moduli in different parts. So when the same amount of stresses are applied, changes in strain are different.

- Cause 3: Specimen state during measurement; Relevant group(s): Almost occur in every specimen.

The specimen is not completely stationary during measurement, which can lead to inaccurate measurement.

- Cause 4: Estimate observing; Relevant group(s): P5; P6; P10; P12; P28; P29; P37; P39; M2; M6; M16; M22; M30; M40

Some groups did not follow the principle of estimate observing, so it is hard to ensure whether the data are accurate and reliable or not, but the exact results after estimate observing cannot be accurately known.

- 2. **Unit analysis:** The original unit of data is Pascal (Pa). When numbers of significant figures are the same, the precision of

data is different. When Pa is used, the precision is very high with low simplicity, when MPa is used, the simplicity is high with low precision. When the unit kPa is used, the precision and simplicity of data recording can both be guaranteed.

3. Data comparison before and after correction of mistakes: After corrections, the calculated modulus moves closer to the average value, and thus lowers the standard deviation and standard error of the sample, making the sample more representative.

Section 4: Investigating whether the data obey the normal distribution.

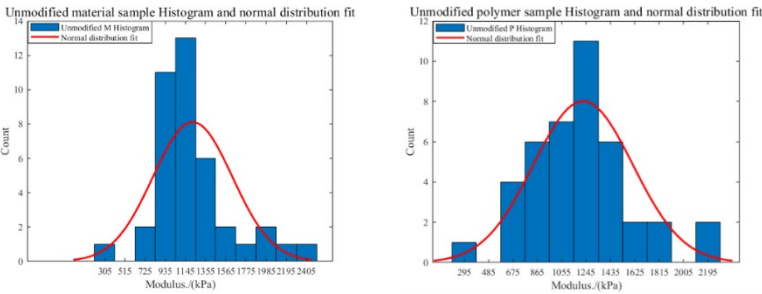
- Method 1: Normality test** Among all testing methods ^{[1][2][3]}, for larger samples (unmodified P&M samples, whose N is around 40), Shapiro-Wilk test would be the most suitable one. While for the small samples (modified P&M samples), Lilliefors test would be the most suitable one. The tests were carried out in Origin and results are summarized in the following Table. 1.

Table. 1 Normality test results using Origin

Samples	Sample size	Test method	Statistic value	p-value	Decision at significance level (5%)
Unmodified Material	40	Shapiro-Wilk	0.91612	0.00583	Reject normality
Unmodified Polymer	41	Shapiro-Wilk	0.97575	0.5195	Can't reject normality
Modified Material	4	Lilliefors	0.2271	0.2	Can't reject normality
Modified Polymer	4	Lilliefors	0.40565	0.02077	Reject normality

At the significance level of 0.05, the data from Unmodified M samples and Modified polymer sample was significantly drawn from a normally distributed population. While the other two did not pass the normality test.

- Method 2: Graphic manner --- Histogram**



As shown in Fig. 1, the histogram for unmodified Material(left) shows that the distribution possesses a slight deviation to the left of the normal distribution fit. While the Polymer sample(right) shows a better fit to the curve which was consistent with the results above in normality test.

Fig. 1 Histogram and normal distribution fit of material (left) and polymer (right) samples

Section 5: Comparison of unmodified P&M samples: Comparable.

- Method 1: Two-sample t-test (2-tailed)** Since the two samples are approximately satisfied normal distribution, with good homogeneity of variance and large enough sample size ($n>30$), the two-sample t-test is selected to examine the significance of difference. Null hypothesis (H_0) states that the means of both samples are the same ($\bar{X}1 = \bar{X}2$), and alternative hypothesis (H_1) refers that the mean of both samples differs ($\bar{X}1 \neq \bar{X}2$). The significance level is set as $\alpha= 0.05$. According to the equation:

$$t = \frac{\bar{X}1 - \bar{X}2}{\sqrt{(S_1^2/n_1) + (S_2^2/n_2)}}$$

the test statistics $d/SE_{(d)}$ can be calculated and $t = 0.015$. The corresponding t-critical value is $t_{(0.975,79)} = 1.990$ for 95% confidence. Since $t < t_{(0.975,79)}$, H_0 is true and this test statistic follows the t distribution. There is no systematic difference between the unmodified M & P samples (two-sample t-test, $p>0.05$).

- Method 2: Graphic** For unmodified M, the central value (mean, μ) is 1220.353 kPa, while for unmodified P the value is 1221.669 kPa, which is slightly different. However, unmodified P sample has more data centered around the central value (justified by the slightly higher peak value of unmodified P than that of unmodified M), indicating the unmodified P sample shows more concentrated distribution than unmodified M sample. The standard deviation(σ) of the two samples also elucidates the same pattern: σ of unmodified P is 387855.3728 Pa, which is smaller than unmodified M ($\sigma = 413213.3347$ Pa).

Section 6: Comparison of modified P&M samples: Incomparable.

Z-test requires either known sample deviation or a large sample ($N>30$), but modified samples did not satisfy both conditions. As a result, these two samples are considered incomparable. If **t-test** is applied, it is required that each sample satisfies normal

[1] Anon, 17.1.7.3 choosing normality tests and interpreting results. OriginLab Corporation - Data Analysis and Graphing Software - 2D graphs, 3D graphs, Contour Plots, Statistical Charts, Data Exploration, Statistics, Curve Fitting, Signal Processing, and Peak Analysis. Available at: <https://www.originlab.com/doc/Origin-Help/NormalityTest-EX> [Accessed January 21, 2022].
[2] Korstanje, J., 2020. 6 ways to test for a normal distribution-which one to use? Medium. Available at: <https://towardsdatascience.com/6-ways-to-test-for-a-normal-distribution-which-one-to-use-9dcf47d8fa93> [Accessed January 21, 2022].
[3] Anon, 17.1.7 normality test. OriginLab Corporation - Data Analysis and Graphing Software - 2D graphs, 3D graphs, Contour Plots, Statistical Charts, Data Exploration, Statistics, Curve Fitting, Signal Processing, and Peak Analysis. Available at: <https://www.originlab.com/doc/Origin-Help/Normality-Test> [Accessed January 21, 2022].

distribution. However, the normality test in section 4 pointed out that modified polymer sample was not drawn from normal distribution. Furthermore, these two samples are too small to calculate for a reliable t-value, so they are incomparable. For rough comparison, the average value of moduli of modified material is larger than that of modified polymer but shows greater standard deviation, which means moduli of modified material possess greater level of dispersion than modified polymer sample. However, due to the lack of data, this comparison is rather weak. More data are needed for a more accurate comparison.

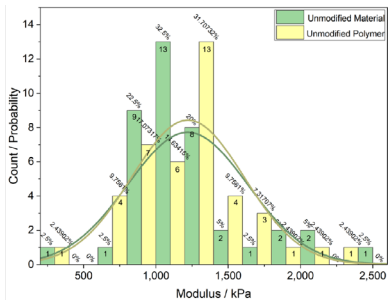


Fig. 2. Normal distribution fit of unmodified M (green) and unmodified P (yellow) samples

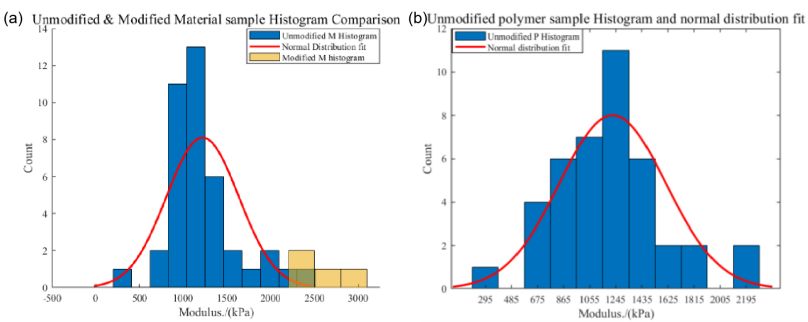


Fig. 3 Histogram of modified and unmodified sample (M(a) and P(b) separately)

Section 7: Comparison of unmodified and modified samples: Comparable.

• Method 1: z-test

The z-test lets us compare whether there is significant difference between new data and a population. Its null Hypothesis (H_0) states that the means of the sample and population are the same. Set significance level $\alpha = 0.05$, critical $z = 1.96$. If $z > 1.96$, null hypothesis is incorrect, it is “statistically significant” that sample and population are different. Otherwise, null hypothesis is correct,

sample and population are the same. Formula for the Z-test is:

$$z = \frac{\bar{X} - \mu}{S_{\bar{X}}}$$

Since the unmodified M&P sample have relatively large sample size ($N \approx 40$, larger than 30), they can be viewed as “population” and normality test in section 4 has confirmed that they were approximately from normal distribution population. Thus, population mean “ μ ” should be the mean values for unmodified P&M samples as sample mean are the unbiased estimator of the population mean [4][5]. The z-scores are calculated in Table. 2.

Table. 2 Calculated z-scores

	$\bar{X}$ (mean of the sample)	μ (Mean of population for comparison)	$S_{\bar{X}}$ (SD of the sample)	z
Material	2528564.5581	1220353.3888	247955.3739	5.2760
Polymer	1974280.651	1221668.7947	75589.4119	9.9566

According to the table, the z-score of both material and polymer samples are larger than 1.96 (Z_{crit}). It is statistically significant that modified sample and population are different for both polymer and material samples.

• Method 2: Graphics

As shown in Fig. 3, the unmodified M&P samples are thought to be approximately normal distribution. Thus, the histogram of the modified and unmodified samples is plotted and compared. It is clearly shown in the graph that modified sample (Yellow) are far away from the mean of unmodified sample distribution (Blue), thus proving the z-test intuitively.

Section 8: Reflection.

Most group members were very devoted to this project, and tasks were divided according to the interests and strength of group members, so there were not many problems with group management, leading to decision making becoming the most important problem. For the solution, before making every important final decision, our group will hold a meeting to discuss the solution, during which process all group members were making contributions, like sharing their ideas, taking notes, or making conclusion, etc., Through this process, each group member was able to share their opinion, so we have a deeper and more comprehensive understanding of the problem. At the same time, we have obtained a variety of methods to solve the problem and learned more knowledge in the thinking collision.

[4] Stuart, Alan; Ord, Keith; Arnold, Steven [F.] (2010). Classical Inference and the Linear Model. Kendall's Advanced Theory of Statistics. 2A. Wiley. ISBN 0-4706-8924-2.
[5] Voinov, Vassily [G.]; Nikulin, Mikhail [S.] (1993). Unbiased estimators and their applications. 1: Univariate case. Dordrecht: Kluwer Academic Publishers. ISBN 0-7923-2382-3.